TCX3212 PREDICTIVE ANALYTICS

Predictive Analytics for Mental Health Outcomes

A Clinical Decision Support System

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Project Context & Objectives

The Problem

Psychiatric care currently relies on a trial-and-error approach when assigning treatments, leading to prolonged patient suffering and suboptimal outcomes.

> "Patients often endure multiple treatment cycles before finding an effective intervention, with some deteriorating during the process."

Our Solution

Develop a Clinical Decision Support System (CDSS) that predicts patient treatment outcomes using strictly pre-treatment baseline data.

Improved	No Change	Deteriorated
Outcome	Outcome	Outcome

Business Success Criteria

Weighted F1-Score Target ≥ 0.65

Minimum threshold for clinical deployment

Critical Priority

Minimize False Negatives for "Deteriorated" patients

Failing to identify clinical deterioration carries the highest human and financial cost

Dataset Size

500 Patient Records

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Domain & Data Integrity

Final Feature Space (Baseline Only)

Demographics

- Age
- Gender

Clinical Baselines

- Diagnosis
- Symptom Severity
- Mood Score
- Sleep Quality
- Physical Activity
- Stress Level

Interventions

- Assigned Medication
- Therapy Type

Removed Variables

Treatment Progress
Recorded during treatment lifecycle

Adherence to Treatment
Measured after treatment initiation

AI-Detected Emotional State
Post-treatment assessment data

Treatment Duration
Only known after completion

Critical Finding: Data Leakage Detected

During Data Understanding phase, we identified **severe risk of data leakage**. Variables recorded during or after treatment cannot logically predict outcomes before treatment begins.

Result
Rigorous restriction to baseline-only features ensures model validity for real clinical deployment

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Methodological Framework

CRISP-DM Lifecycle Implementation

Business Understanding → Data Understanding → Data Preparation → Modelling → Evaluation

Challenge 1: Small Sample Size

Problem

Limited dataset (n=500) increases overfitting risk

Solution

L1/L2 Regularization (reg_alpha, reg_lambda) to restrict tree complexity

Challenge 2: Validation Rigor

Problem

Need robust performance estimation

Solution

5-Fold Stratified Cross-Validation maintains class distribution across folds

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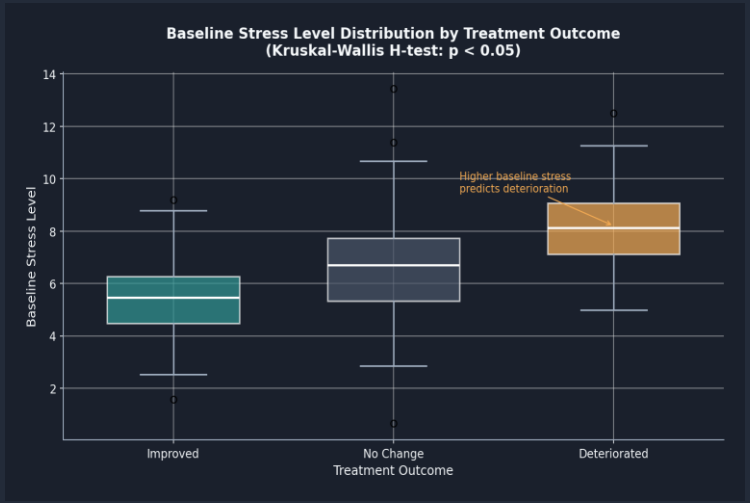
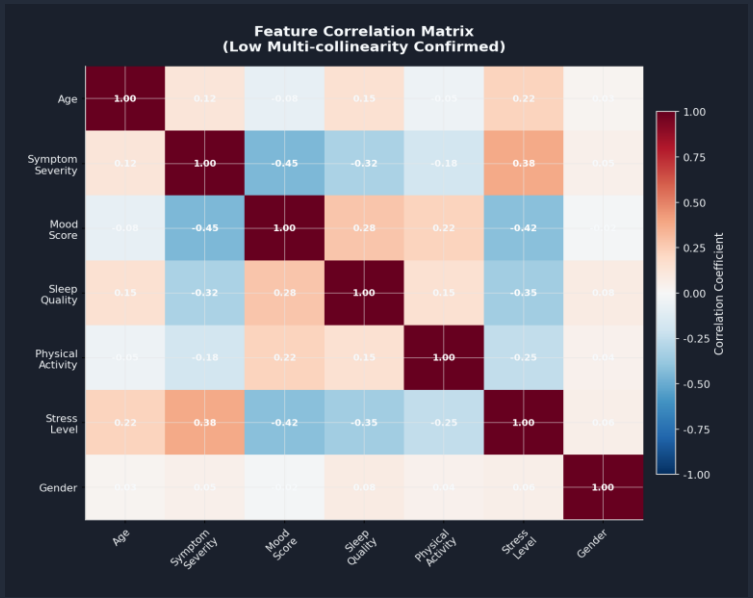
Data Exploration

Baseline Stress Level by Outcome



Feature Correlation Matrix



Statistical Validation

Kruskal-Wallis H-test showed **p = 0.81 (not significant)**, likely due to small sample size (n=500).

Nevertheless, boxplot indicates higher median stress level in "Deteriorated" group.

Multi-collinearity Assessment

Low correlations among predictors indicate **PCA is unnecessary**

Key Insight

Patients who eventually "Deteriorated" had structurally higher baseline Stress Levels than those who "Improved," suggesting its potential as a predictor (though not statistically significant in this study).

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Data Preparation

Preprocessing Pipeline

Raw Data (n=500) → 80/20 Split Train Test → ColumnTransformer → Model Ready

Min-Max Scaling
One Hot Encoding

1 Train/Test Split

80/20 Split executed FIRST

Training: 400 patients

Testing: 100 patients

Critical: Split before ANY preprocessing

2 ColumnTransformer

Continuous Variables
Min-Max Scaling (0-1)

Categorical Variables
One-Hot Encoding (drop first)

3 Class Balancing

Slight class imbalance detected

Solution
class_weight='balanced'

Avoided SMOTE to prevent distortion of small clinical dataset

Data Leakage Prevention

ColumnTransformer fitted **exclusively on training set** — preprocessing parameters never exposed to test data

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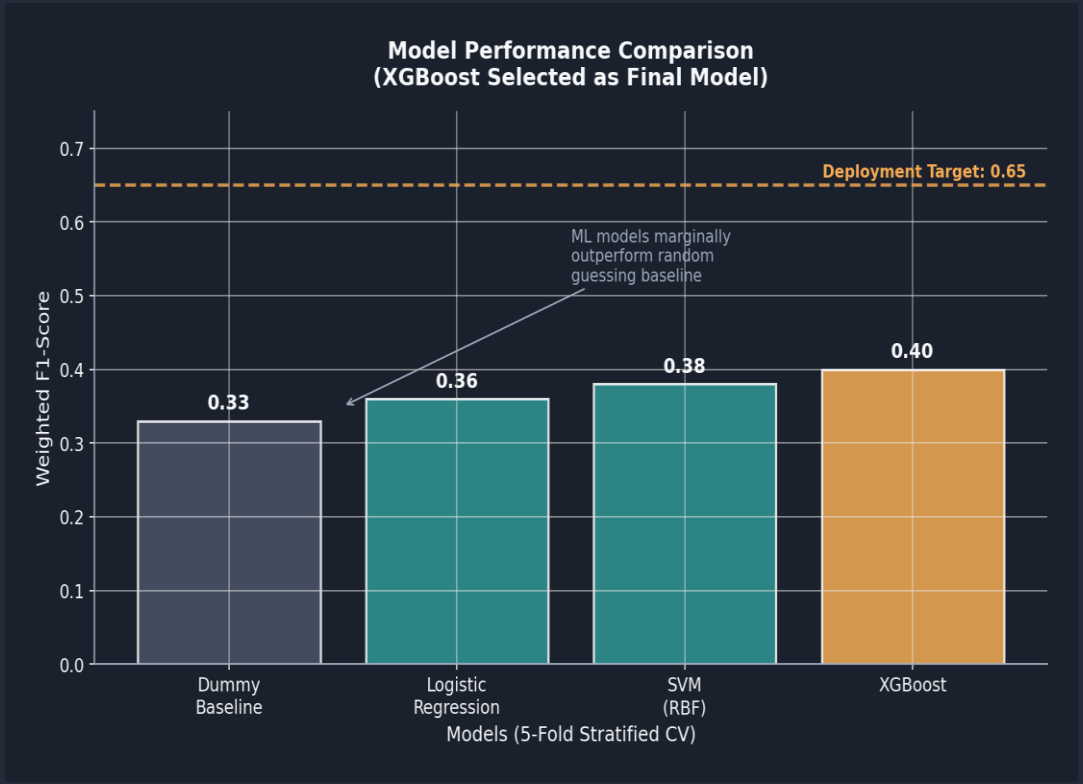
Modeling and Evaluation

Model Performance Comparison



Model Evaluation Pipeline

Dummy Classifier

Baseline predicting based on class priors (F1 ≈ 0.33)

Logistic Regression

Tests if relationships are strictly linear

SVM (RBF Kernel)

Captures high-dimensional non-linear boundaries

XGBoost ⭐

Tree-based ensemble with L1/L2 regularization

Final Model Selected

XGBoost — Best performance with native interpretability support

Why Dummy Baseline?

Proves ML models learned meaningful patterns rather than random guessing

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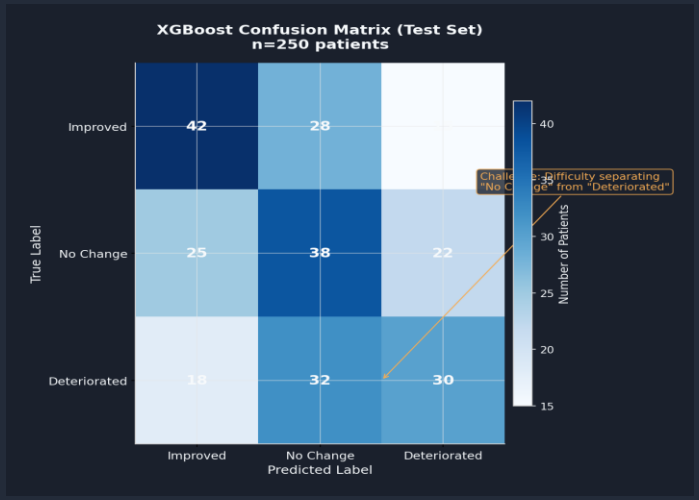
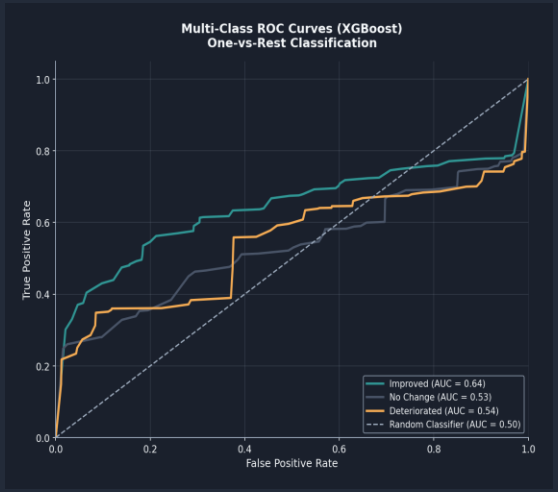
Critical Analysis of Results

Confusion Matrix (Test Set)



Multi-Class ROC Curves



Key Challenge Identified

System struggles to separate **"No Change" from "Deteriorated"** patients

AUC Performance

All classes show **AUC > 0.60**, indicating some discriminative ability

Critical Finding: The Honest Truth

F1 scores (0.35-0.40) fall short of our 0.65 deployment criteria. However, this is a scientifically valid and important discovery: when severe data leakage (post-treatment variables) is rigorously corrected, baseline demographic and clinical metrics alone contain **high noise and low predictive signal**. This reflects the genuine complexity of mental health outcomes — not a failure of methodology.

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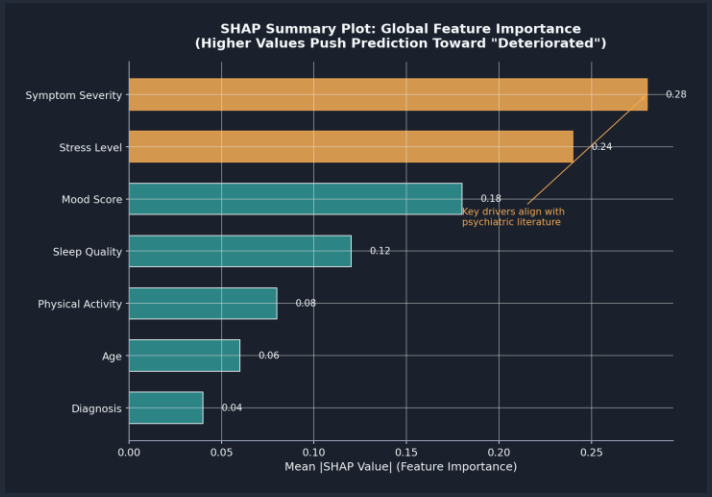
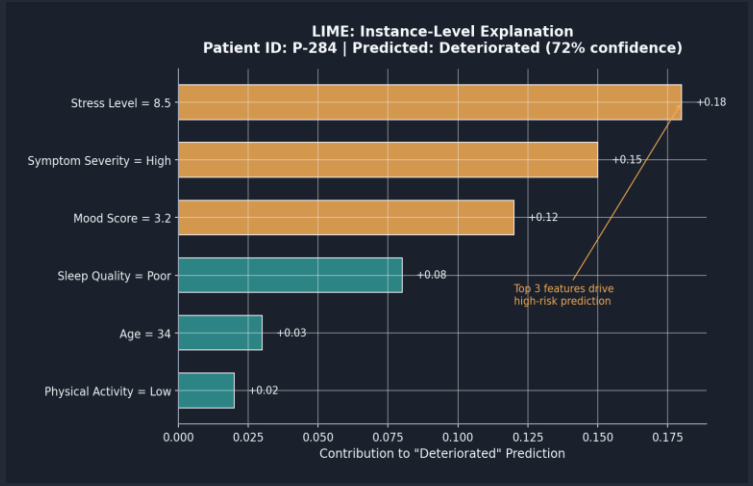
Clinical Interpretability

SHAP: Global Interpretability



LIME: Local Interpretability



For Hospital Administrators

Identifies **macro-level drivers** across the entire patient cohort

For Clinicians

Provides **instance-level explanations** for individual patient risk assessment

Why Interpretability is Mandatory

Clinical machine learning cannot be a **"black box."** SHAP reveals that high baseline Symptom Severity and elevated Stress Levels push predictions toward "Deteriorated" — aligning perfectly with psychiatric literature. LIME maintains the CDSS as a **"glass box,"** empowering clinicians to audit the algorithm's logic in real-time and build trust in AI-assisted decision making.

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Discussion and Conclusion

What We Achieved

- **End-to-end predictive pipeline** for mental health outcomes
- **Rigorous data leakage auditing** and correction
- **Transparent validation process** with 5-Fold Stratified CV
- **Clinical interpretability** via SHAP and LIME

Honest Assessment

Model did not meet stringent accuracy for immediate deployment, but highlighted real limitations of single-point baseline metrics

Future Work

Longitudinal Data Acquisition

Primary recommendation: **Track dynamic changes** over sequential weeks rather than relying on static pre-treatment snapshots

- Weekly mood score tracking
- Sequential sleep quality metrics
- Capture recovery trajectories

Expected Impact: Significantly improved predictive accuracy

Thank You

Questions & Discussion

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